(Т1) $\xi \sim R(0, \theta)$, $\theta > 0$

$\theta \in \Theta$, $\Theta = (0, \infty)$

выборка $\vec{x}_n$

$\bar{x} = 2 \cdot \frac{1}{n} \sum_{i=1}^{n} x_i$

$\tilde{\theta}_1 = 2\bar{x}$, $\tilde{\theta}_2 = X_{min}$, $\tilde{\theta}_3 = X_{max}$, $\tilde{\theta}_5 = \left(x_1 + \frac{1}{n-1} \sum_{k=2}^{n} x_k\right)$

Плотность равном распред $p(x, \theta) = \frac{1}{\theta} \{(0, \theta)\}$

$M\xi = \int_0^{\theta} x \cdot \frac{1}{\theta} dx = \frac{\theta}{2}$

$M\xi^2 = \int_0^{\theta} x^2 \cdot \frac{1}{\theta} dx = \frac{\theta^2}{3}$

$D\xi = M\xi^2 - (M\xi)^2 = \frac{\theta^2}{3} - \frac{\theta^2}{4} = \frac{\theta^2}{12}$

а) Проверить на несмещенность и состоятельность

(1) $\tilde{\theta}_1$: $\forall \theta > 0$ $M\tilde{\theta}_1 = M\left(2 \cdot \frac{1}{n} \sum_{i=1}^{n} x_i\right) = \frac{2}{n} \cdot \sum_{i=1}^{n} Mx_i = \frac{2}{n} \cdot n M\xi =$

$= \theta$ – несмещенная

$D\tilde{\theta}_1 = D\left(2 \cdot \frac{1}{n} \sum_{i=1}^{n} x_i\right) = \frac{4}{n^2} \sum_{i=1}^{n} Dx_i = \frac{4}{n^2} \cdot n D\xi = \frac{\theta^2}{3n}$

$\frac{\theta^2}{3n} \xrightarrow[n \to \infty]{} 0$ $\forall \theta > 0$ => оценка состоятельна по дост. условию

② $\tilde{\theta}_2$: $\forall \theta > 0$ $\quad M\tilde{\theta}_2 = M x_{min}$

$\xi \sim F(x) \quad \Rightarrow \quad \xi_{min} \sim \underbrace{1-(1-F(x))^n}_{\Phi(x)}$

$$\Phi(x) = \Phi'(x) = n(1-F(x))^{n-1} F'(x) = n(1-F(x))^{n-1} F'(x) =$$

$$= n\left(1-\frac{x}{\theta}\right)^{n-1} \cdot \frac{1}{\theta} \quad \{(0,\theta)\}$$

$$M x_{min} = \int_0^\theta x \cdot n \Big(\underbrace{1-\frac{x}{\theta}}_{t}\Big)^{n-1} \frac{1}{\theta}\, dx = -\int_1^0 n\theta(1-t)t^{n-1}\theta\, dt =$$

$$= n\theta \int_0^1 (t^{n-1} - t^n)\, dt = n\theta\left(\frac{1}{n} - \frac{1}{n+1}\right) = \frac{\theta}{n+1}$$

$\Rightarrow$ оценка смещенная

Возьмем $\tilde{\theta}_2' = (n+1) x_{min}$

$M\tilde{\theta}_2' = (n+1) M x_{min} = \theta$ – несмещенная

$$D\tilde{\theta}_2' = D((n+1)x_{min}) = (n+1)^2 D x_{min}$$

$$M x_{min}^2 = \int_0^\theta x^2 n \Big(\underbrace{1-\frac{x}{\theta}}_{t}\Big)^{n-1} \cdot \frac{1}{\theta}\, dx = \int_0^1 \theta^2 (1-t)^2 n t^{n-1}\, dt =$$

$$= n\theta^2 \int_0^1 (t^{n-1} - 2t^n + t^{n+1})\, dt = n\theta^2\left(\frac{1}{n} - \frac{2}{n+1} + \frac{1}{n+2}\right) =$$

$$= \frac{2\cdot\theta^2}{(n+1)(n+2)}$$

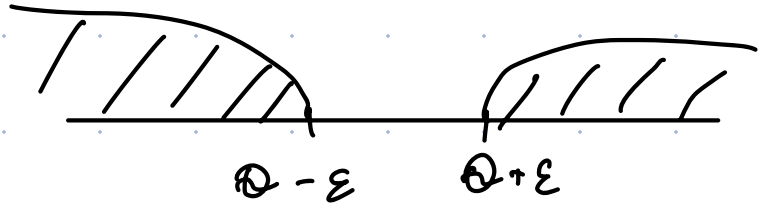
$$D x_{min} = \frac{2\theta^2}{(n+1)(n+2)} - \frac{\theta^2}{(n+1)^2} = \frac{n\theta^2}{(n+1)^2(n+2)} =$$

$$= \theta^2\left(\frac{2(n+1)-(n+2)}{(n+1)^2\,(n+2)}\right) = \theta^2\,\frac{n}{(n+1)^2(n+2)}$$

$$D\tilde{\theta}_2' = D((n+1)X_{min}) = (n+1)^2 D X_{min} = \frac{n\theta^2}{n+2} \underset{n\to\infty}{\not\to} 0$$

достаточное усл. не работает

По опр. $\forall \theta > 0 \quad \forall \varepsilon > 0 \quad P(|\tilde{\theta}_2' - \theta| \geq \varepsilon) \xrightarrow[n\to\infty]{} 0$



$$P(|\tilde{\theta}_2' - \theta| \geq \varepsilon) \geq$$

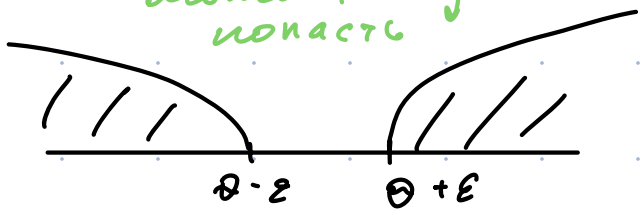
$$\geq P(\tilde{\theta}_2' \geq \theta + \varepsilon) =$$

$$= P((n+1)X_{min} \geq \theta + \varepsilon) = P\left(X_{min} \geq \frac{\theta+\varepsilon}{n+1}\right) =$$

$$= 1 - \underbrace{\Phi\left(\frac{\theta+\varepsilon}{n+1}\right)}_{1-(1-F(x))^n} = 1 - \left(1 - \left(1 - \frac{\theta+\varepsilon}{\theta(n+1)}\right)^n\right) =$$

$$= \left(1 - \frac{\theta+\varepsilon}{\theta(n+1)}\right)^n \xrightarrow[n\to\infty]{} e^{-\frac{\theta+\varepsilon}{\theta}} > 0$$

сюда не можем попасть



Состоятельна ли $\tilde{\theta}_2$

$$P(|X_{min} - \theta| \geq \varepsilon) =$$

$$= P(X_{min} \leq \theta - \varepsilon) = \Phi(\theta - \varepsilon) = 1 - (1 - F(\theta - \varepsilon))^n =$$

$$= 1 - \left(1 - \frac{\theta - \varepsilon}{\theta}\right)^n = 1 - \left(\frac{\varepsilon}{\theta}\right)^n \xrightarrow[n\to\infty]{} 1$$

$\tilde{\theta}_2$ тоже не состоят.

3 $\tilde{\theta}_3 = x_{max}$

$M\tilde{\theta}_3 = M x_{max}$

$x_{max} \sim F(x)^n$

$\varphi(x) = \psi(x) = n\left(\frac{x}{\theta}\right)^{n-1} \frac{1}{\theta} \quad \{[0,\theta)\}$

$$M x_{max} = \int_0^\theta x \cdot n \left(\frac{x}{\theta}\right)^{n-1} \frac{1}{\theta}\, dx = \frac{n}{\theta^n} \frac{\theta^{n+1}}{n+1} = \frac{n}{n+1}\theta$$

смещенная

Возьмем $\tilde{\theta}_3' = \frac{n+1}{n} x_{max}$

$M\tilde{\theta}_3' = \frac{n+1}{n} M x_{max} = \theta$ несмещенная

$$D\tilde{\theta}_3' = \left(\frac{n+1}{n}\right)^2 D x_{max}$$

$$M x_{max}^2 = \int_0^\theta x^2 \cdot n \left(\frac{x}{\theta}\right)^{n-1} \cdot \frac{1}{\theta}\, dx = \frac{n}{\theta^n} \cdot \frac{\theta^{n+2}}{n+2} = \frac{\theta^2 n}{n+2}$$

$$D x_{max} = \theta^2\left(\frac{n}{n+2} - \frac{n^2}{(n+1)^2}\right) = \theta^2\left(\frac{n(n+1)^2 - n^2(n+2)}{(n+2)(n+1)^2}\right) =$$

$$= \frac{n\theta^2}{(n+1)^2(n+2)}$$

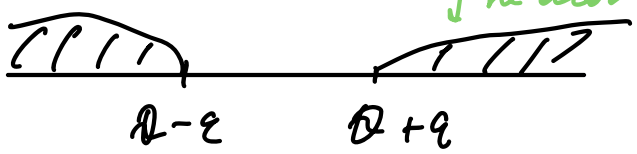
$$D\tilde{\theta}_3' = \left(\frac{n+1}{n}\right)^2 D\tilde{\theta}_3 = \frac{(n+1)^2}{n^2} \cdot \frac{n\theta^2}{(n+1)^2(n+2)} = \frac{\theta^2}{n(n+2)} \xrightarrow[n\to\infty]{} 0$$

состоятельная

Докажем по определению:

$\tilde{\theta}_3'$: $\forall \theta > 0 \ \forall \varepsilon > 0 \quad P(|\tilde{\theta}_3' - \theta| \geq \varepsilon) \xrightarrow[n \to \infty]{} 0$

сюда попасть не можем



$P(\tilde{\theta}_3' \geq \theta + \varepsilon) = 0$, т.к.

$$\tilde{\theta}_3' = \frac{n+1}{n} X_{max} \leq \frac{n+1}{n}\theta = \theta + \frac{\theta}{n}$$

$\Rightarrow$ если $n > \frac{\theta}{\varepsilon}$, то $\frac{\theta}{n} < \varepsilon \Rightarrow$

$\Rightarrow \theta + \frac{\theta}{n} < \theta + \varepsilon$

$$P(\tilde{\theta}_3' \leq \theta - \varepsilon) = P\left(\left(\frac{n+1}{n}\right) \cdot X_{max} \leq \theta - \varepsilon\right) =$$

$$= P\left(X_{max} \leq \frac{n}{n+1}(\theta - \varepsilon)\right) = \left(\frac{n}{n+1} \cdot \frac{(\theta - \varepsilon)}{\theta}\right)^n =$$

$$= \left(\frac{n}{n+1} \cdot \frac{\theta - \varepsilon}{\theta}\right)^n = \left(\frac{n}{n+1} \cdot \left(1 - \frac{\varepsilon}{\theta}\right)\right)^n \leq$$

$0 < 1 - \frac{\varepsilon}{\theta} < 1 \qquad 0 < \frac{n}{n+1} < 1$

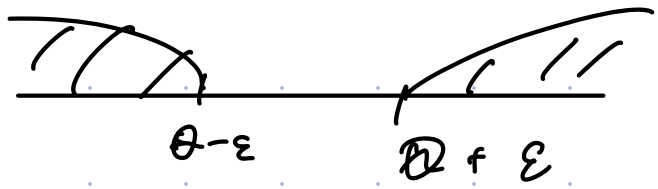
$0 < \frac{n}{n+1}\left(1 - \frac{\varepsilon}{\theta}\right) \leq 1 - \frac{\varepsilon}{\theta} < 1$

$\leq \left(1 - \frac{\varepsilon}{\theta}\right)^n \xrightarrow[n \to \infty]{} 0 \Rightarrow$ оценка состоятельная

Теперь $\tilde{\theta}_3$

$P(|X_{max} - \theta| \geq \varepsilon) =$

$P(X_{max} \geq \theta + \varepsilon) = 0$



$$P(X_{max} \leq \theta - \varepsilon) = (F(\theta - \varepsilon))^n = \left(\frac{\theta - \varepsilon}{\theta}\right)^n =$$

$$= \left(1 - \frac{\varepsilon}{\theta}\right)^n \xrightarrow[n \to \infty]{} 0 \quad \Rightarrow \quad \tilde{\theta}_3 \text{ - состоятельна}$$

p) $\tilde{\theta}_4 = X_1 + \frac{1}{(n-1)} \sum_{i=2}^{n} X_i$

несмещенная

$$M\tilde{\theta}_4 = MX_1 + \frac{1}{n-1} \sum_{i=2}^{n} MX_i = M\xi + \frac{1}{n-1} \sum_{i=2}^{n} M\xi = \frac{\theta}{2} + \frac{\theta}{2} = \theta$$

$$D\tilde{\theta}_4 = DX_1 + \frac{1}{(n-1)^2} \sum_{i=2}^{n} DX_i = \frac{\theta^2}{12} + \frac{1}{(n-1)} \frac{\theta^2}{12} \not\to 0$$

не вып. дост. усл

ξ_n - незав. и одинаково распределены, мат. ожидания конечны $\Rightarrow$ по ЗБЧ Хинчина

$$\frac{1}{n} \sum_{i=1}^{n} \xi_n \xrightarrow{P} M\xi_1$$

$$X_1 + \frac{1}{n-1} \sum_{i=2}^{n} X_i \xrightarrow{P} \xi + \frac{\theta}{2} \Rightarrow$$ не явл. состоят.

б) Эффективность

$\tilde{\theta}_3' = \frac{n+1}{n} x_{max}$ $\qquad D\tilde{\theta}_3' = \frac{\theta^2}{(n+2)n}$

$\tilde{\theta}_1 = 2\bar{x}$ $\qquad D\tilde{\theta}_1 = \frac{\theta^2}{3n}$

Сравним $\frac{1}{3n}$ и $\frac{1}{n(n+2)}$

$3n < n^2 + 2n$

$n > 1$

$\tilde{\theta}_3'$ более эффективна, чем $\tilde{\theta}_1$ при $n > 1$